Industry Life Cycle & Competition in Mature Industries

Session 18

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Why is it relevant to take into account the industry life cycle?

Things change over time



1. The industry life cycle

2. Characteristics of each stage of the life cycle

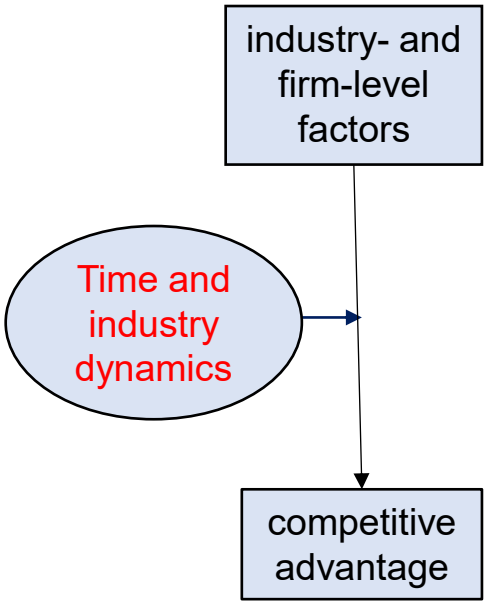
3. Life Cycle Industry Dynamics

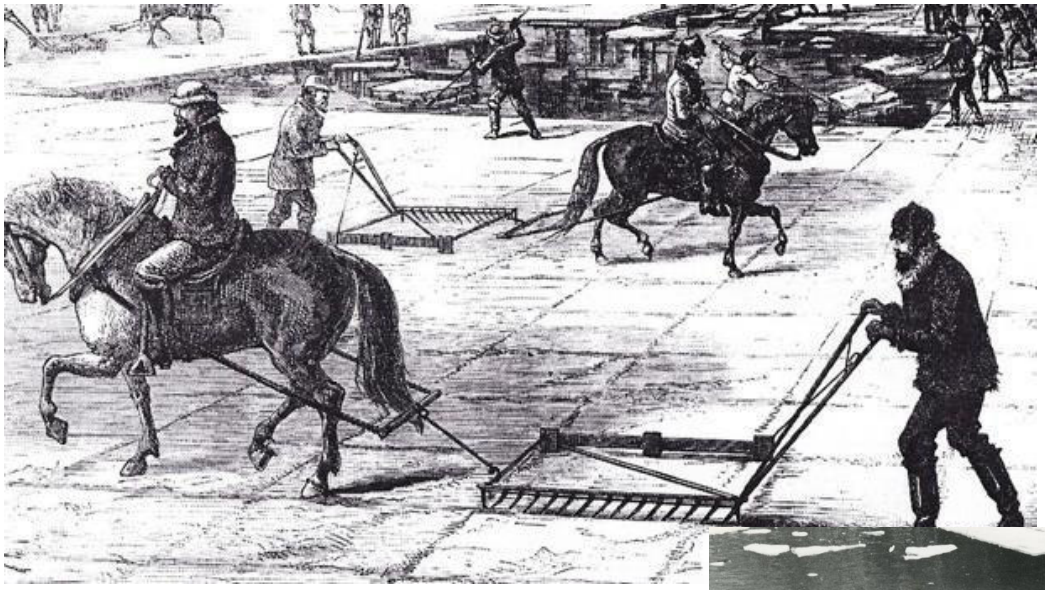
4. Different life cycle Patterns

5. Mature industries: Competitive advantage

Why is it relevant the industry life cycle?

1. The Industry environment is in a state of constant change.
2. Throughout the industry life cycle new firms are born, existing firms adapt or die.
3. We need to understand the factors that drive change in that industry. Change is driven by:
   - Technology / diffusion of knowledge.
     Ex. Internet, electric car
   - Consumer needs / demand
     Ex. Smartphone, PC
   - Economic development
     Ex. Development of great countries as: China and India
   - Exogenous shocks. Ex. Covid, wars, etc..





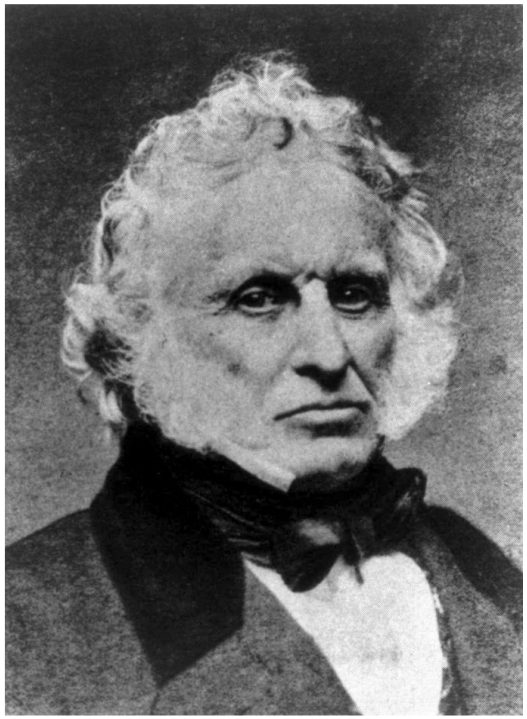


What are they doing?

The ice harvesting industry



- Highly profitable industry in the 19th century
- In 1856, a company was shipping 146,000 tons of ice

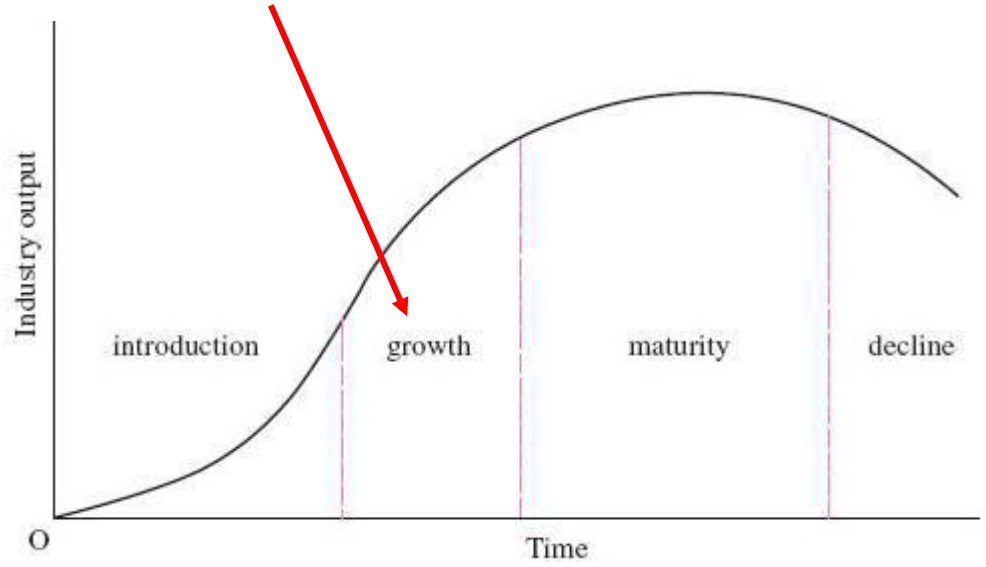


Frederic Tudor, the Ice King

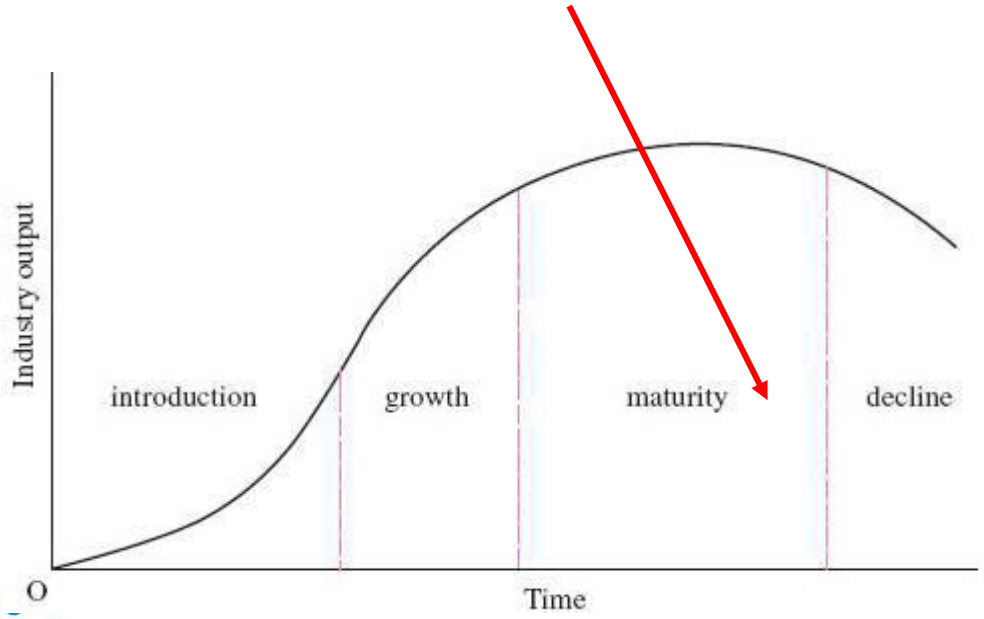
Over time:

The industry became more and more competitive

Major innovations and new methods: Ice harvesting got more efficient (uniform blocks of ice could easily be transported and stored up to 3 years!)



- Ice harvesters developed an entire system for production, storage, and distribution that was remarkably efficient
- Late 19th century: Rapid spread of machine that makes ice (REFRIGERATOR / FRIDGE)
- The industry declined eventually



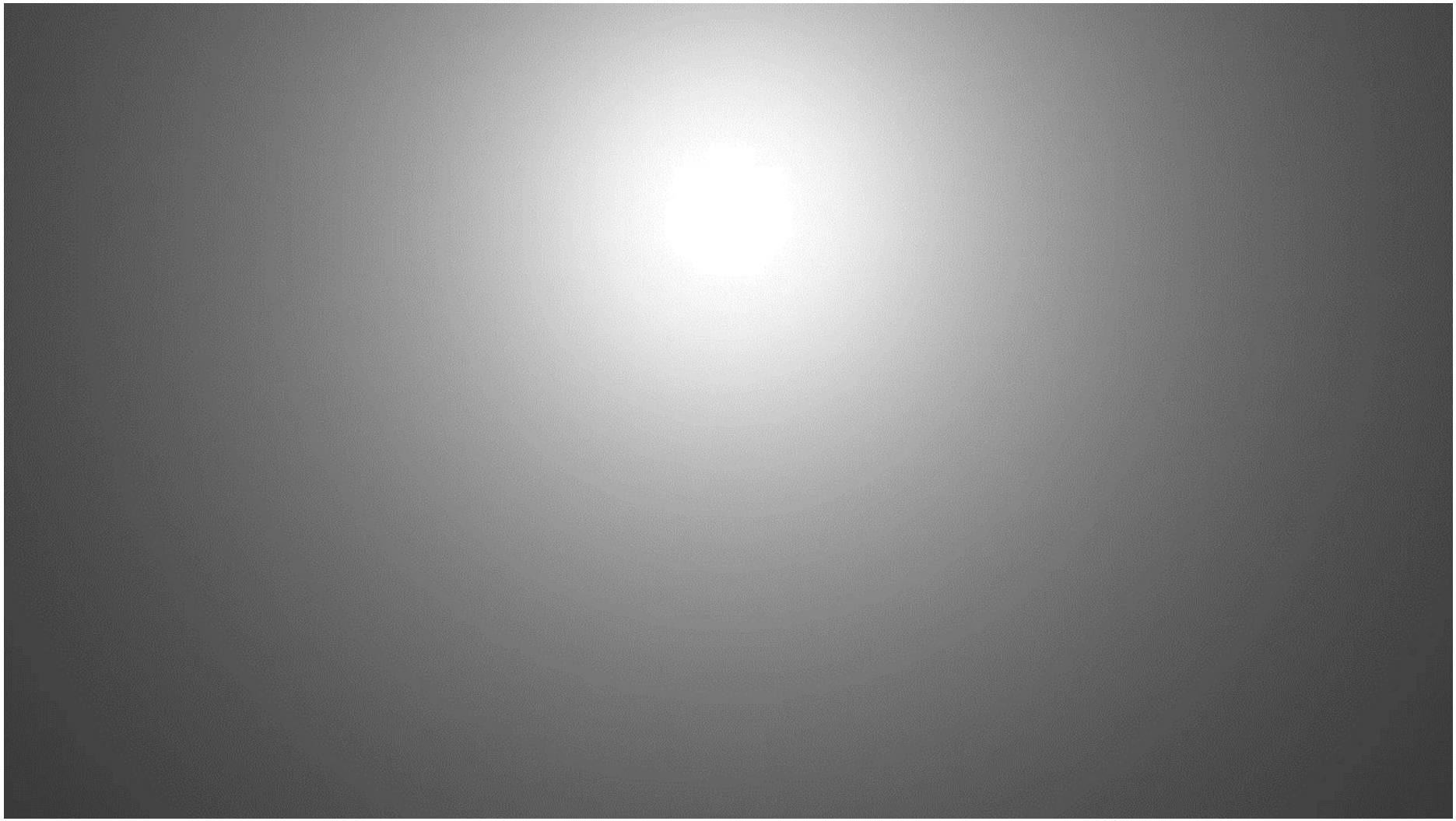


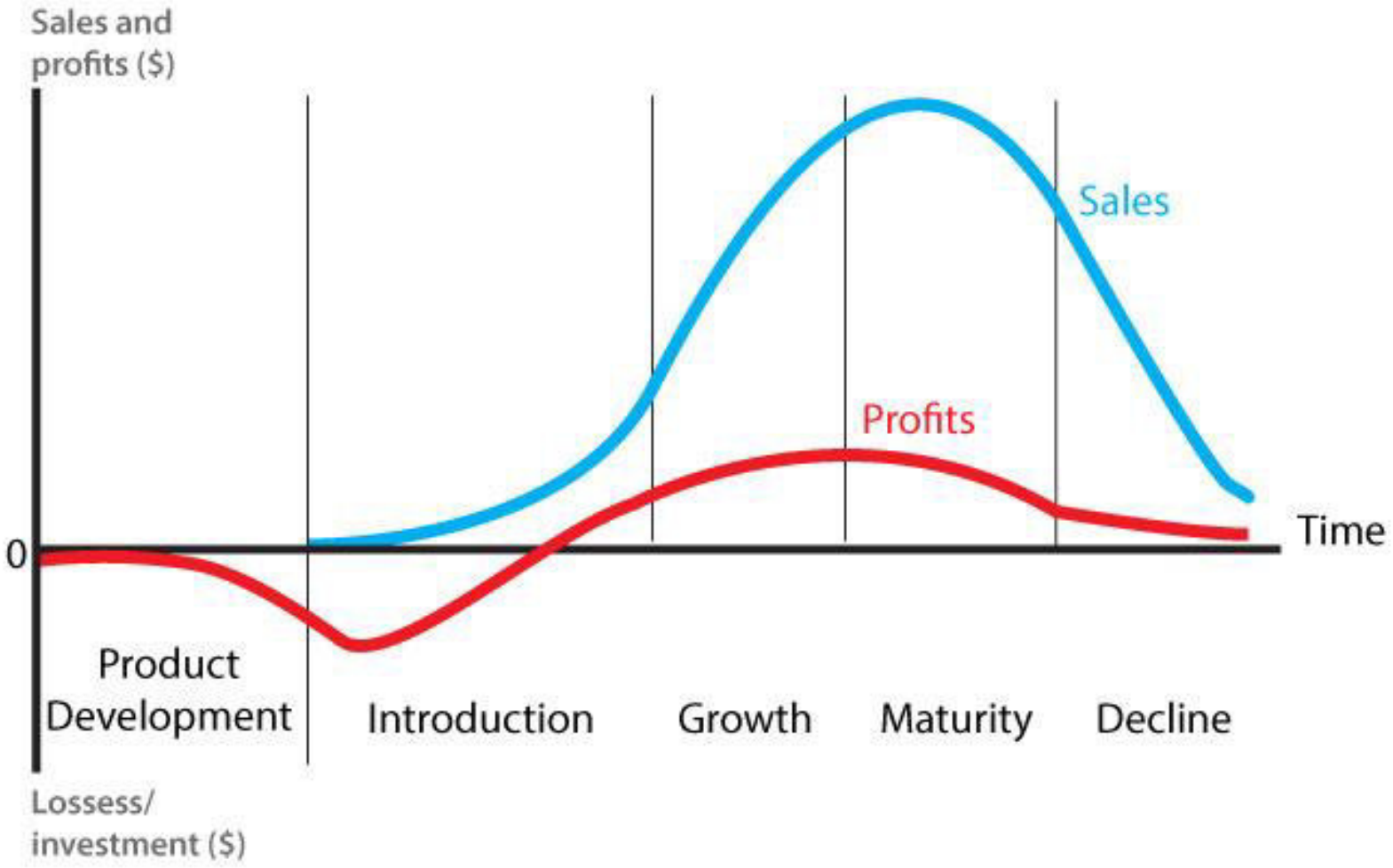
Despite the rapid spread of machine-made ice in 1886, ice harvesting was strong as ever–25 million tons

Threat of technology can be obscured by a growing market!

What is INDUSTRY life cycle? (4m.)

https://www.youtube.com/watch?v=e-P1RMTOedE




Sales and profits ($)
Sales
Profits
0
Time
Product Development
Introduction
Growth
Maturity
Decline
Lossess/ investment ($)

1. The industry life cycle

2. Characteristics of each stage of the life cycle

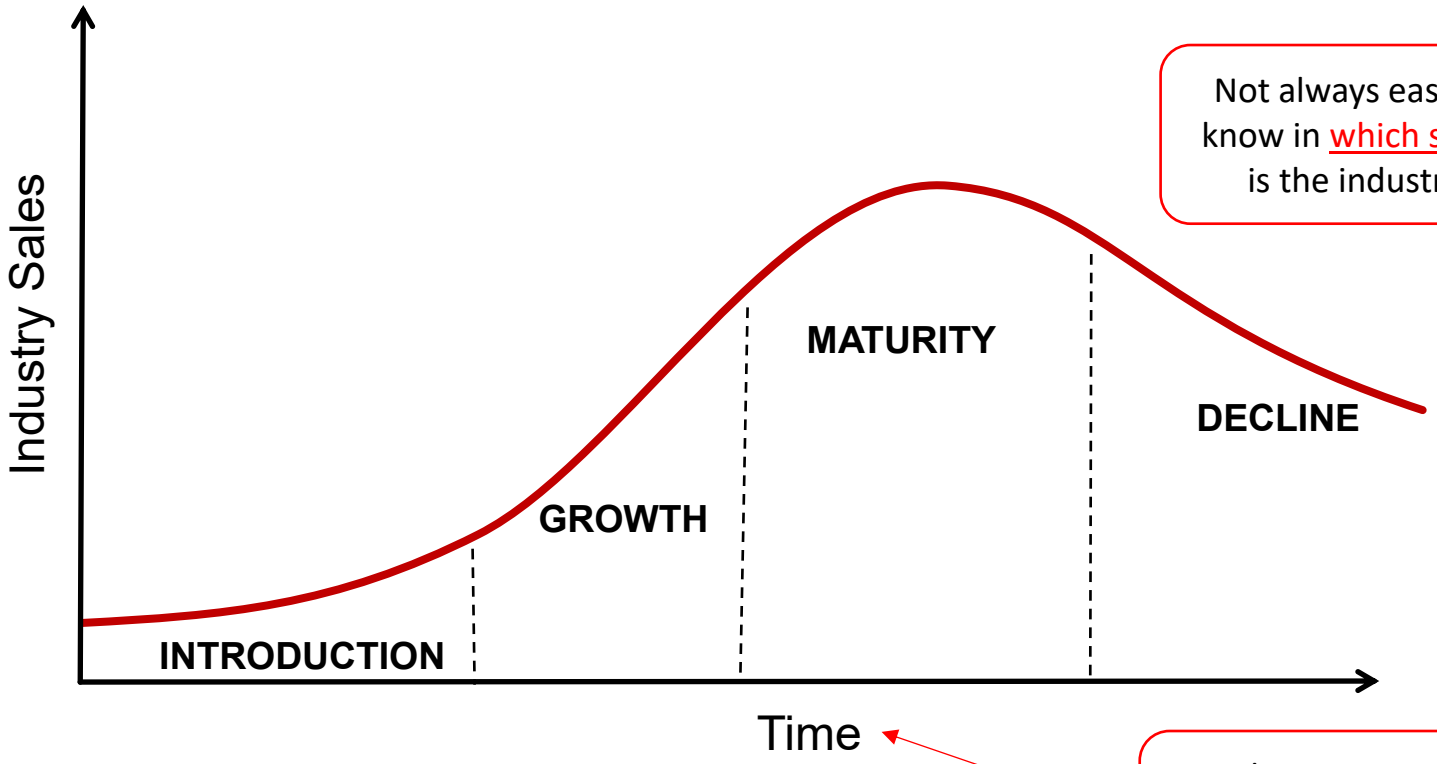
3. Life Cycle Industry Dynamics

4. Different life cycle patterns

5. Competitive advantage in mature industries

Stages of the Industry Life Cycle

Not easy to know the sales. Industry combine different products and several segments and competitive positions



Not always easy to know in which stage is the industry

Not always easy to know which is the time line of the industry

Drivers of industry evolution :

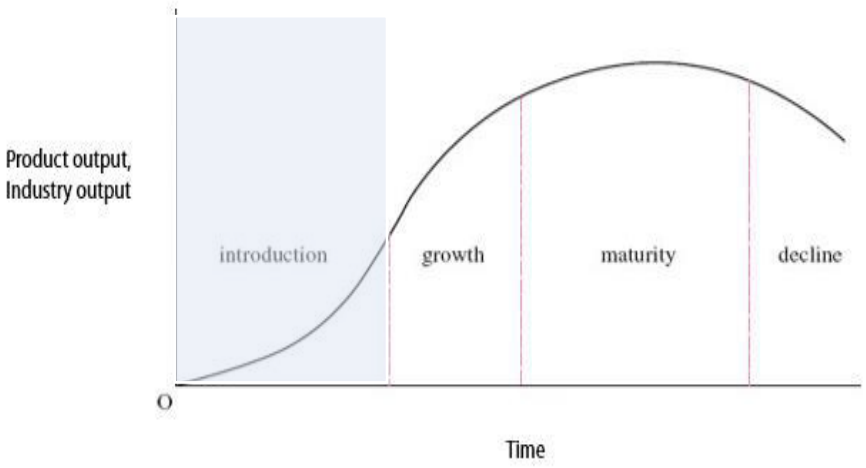
1. **DEMAND growth**
2. **Creation + Diffusion of KNOWLEDGE**

1. Introduction

phase of the industry lifecycle



1. Introduction phase of the industry lifecycle



Demand

- **Sales**: Low sales and low rate or market penetration
- **Customers**: innovation-oriented early-adopters
- **Production**: low scale, expensive, low quality

Diffusion of knowledge

- New knowledge in the form of Product innovation is responsible for an industry's birth
- No dominant product or technology. Product technology advances rapidly
- Competition among designs

Dominant design vs. Technical Standard

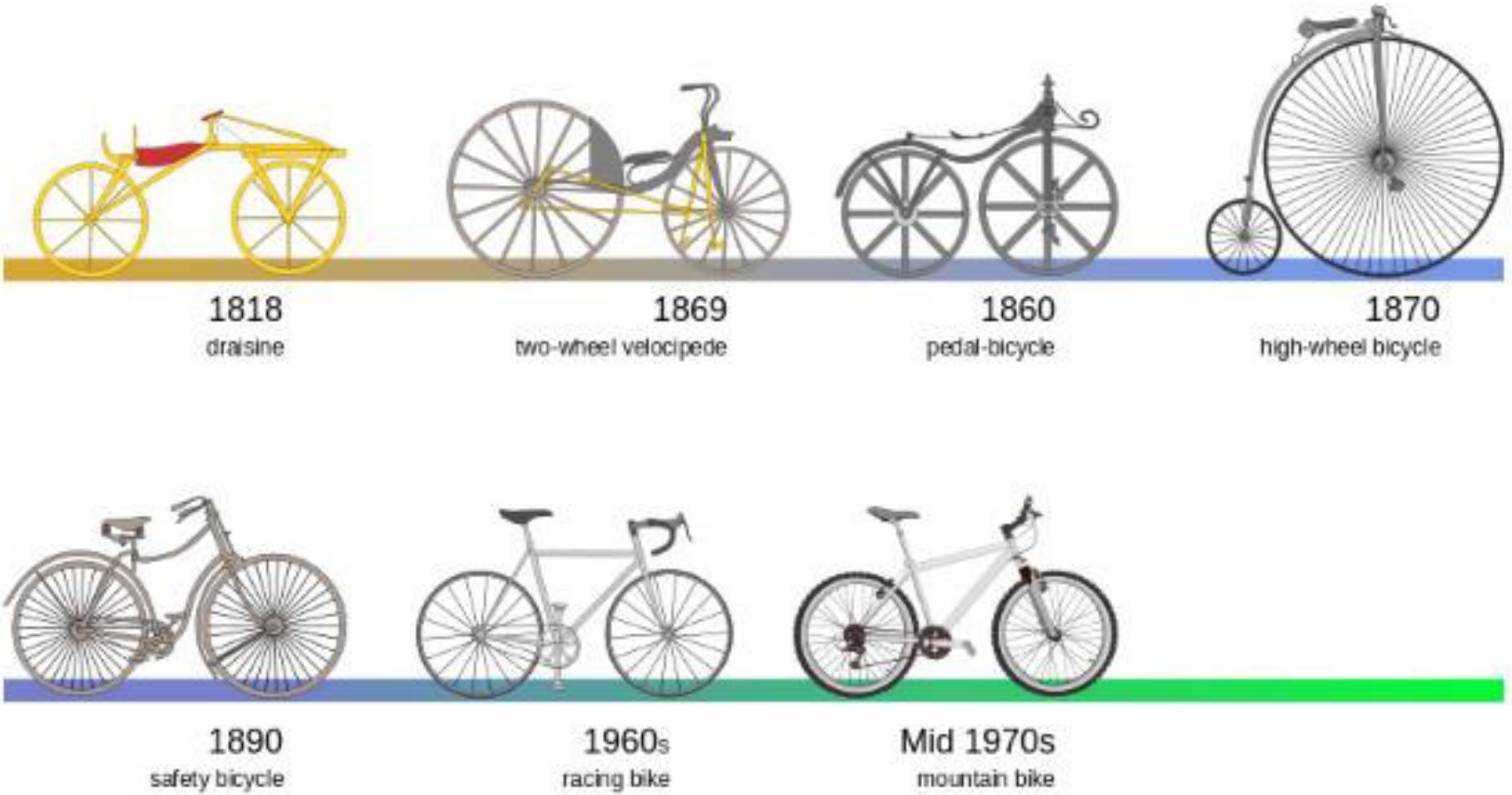
DOMINANT DESIGN → Product architecture that defines the look, functionality, and production method for the product and becomes accepted by the industry as a whole. (e.g.: touchpad smartphones; excel type format; message apps)

- Dominant design ≠ (not always) technical standard
- A dominant design may or may not embody a technical standard

TECHNICAL STANDARD → Technology or specification that is important for **compatibility** (e.g. Android vs. iOS Apple; PDF format; MS Office vs. open Office; WhatsApp or Telegram)

- Technical standards typically embody intellectual property in the form of patents or copyrights, a dominant design usually does not.
- Technical standards especially emerge when there are network effects (consumers benefit from using the same standard) (e.g. Andoid vs IOS / Whatsapp vs. Telegram)

Dominant design of the bicycle



Examples of technical standards



Ethernet

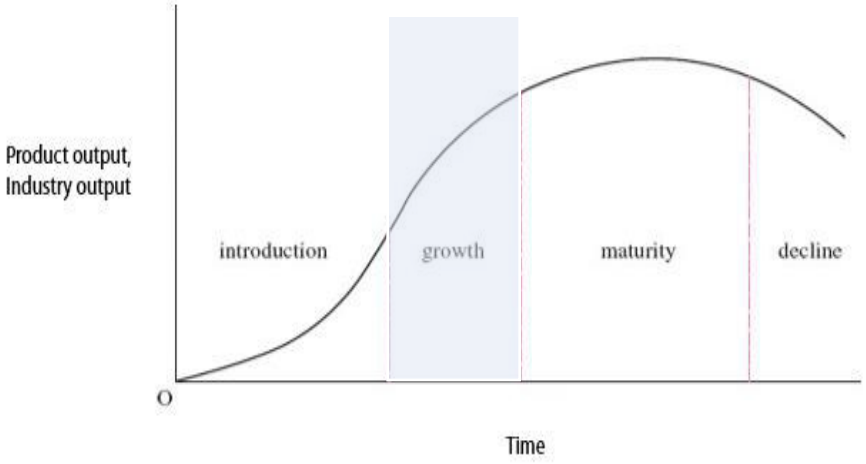


USB-C



5G

2. Growth phase of the industry lifecycle



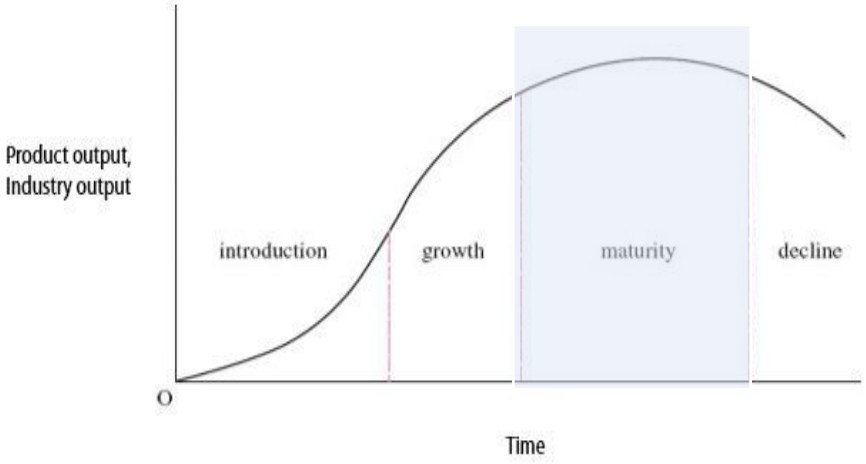
Demand

- **Sales**: Accelerated sales and market penetration.
- **Customers**: transition to mass-market
- **Production**: technical improvements and increased efficiency (lower costs, higher quality)
- More firms enter, first mergers and exits – more competition among firms

Diffusion of knowledge

- Emergence of a dominant design
- Increased standardization

3. Maturity phase of the industry lifecycle



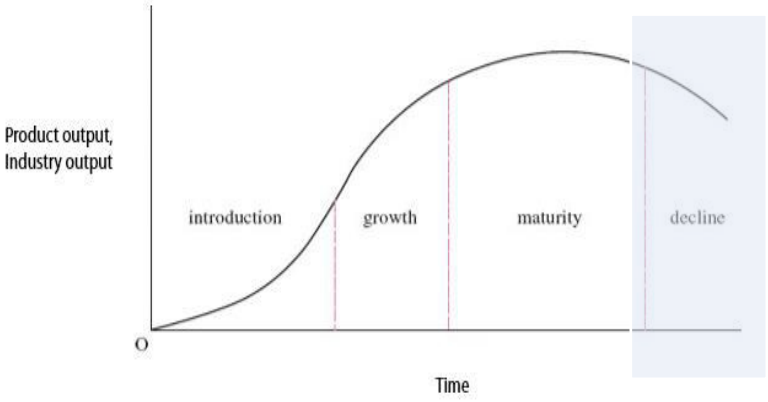
Demand

- **Sales**: Market saturation
- **Customers**: increasingly price sensitive
- **Production**: quest for process improvements, emergence of overcapacity
- **Competitors**: Large firms
- **Price**: Shakeout and price competition increase

Diffusion of **knowledge**

- Refinement of the dominant design
- Increasing differentiation by branding, quality, and bundling

4. Decline phase of the industry lifecycle



Demand

- **Substitute products**: Industry becomes challenged by new industries that produce technologically superior substitute products
- **Sales**: Declining sales
- **Customer**: highly price sensitive
- **Production**: overcapacity is common
- **Price**: Firms exit, price wars are common

Diffusion of knowledge

- Little innovation
- Dominant design substituted by new technologies

Innovation over the industry lifecycle





Internal combustion



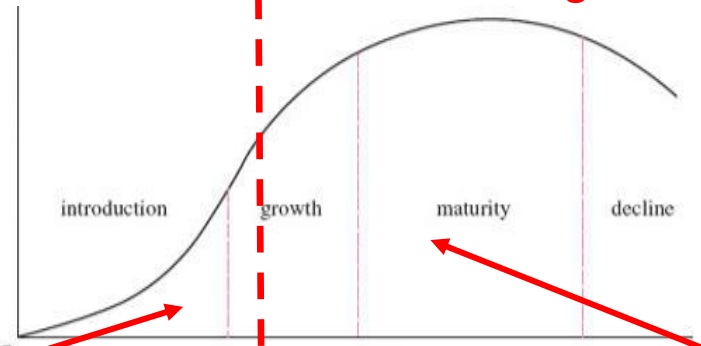
Ford Model T, **1908**

- front engine
- internal combustion
- water cooled

Dominant Design



Steam

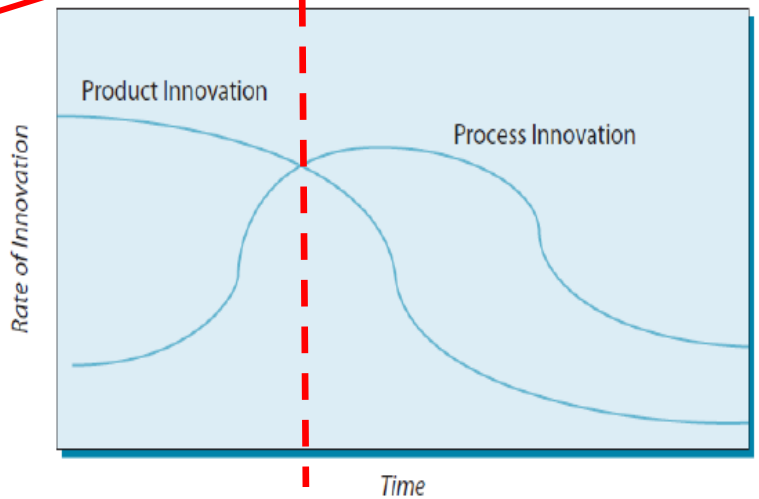




Ford Highland Park
Assembly Plant

Early phase

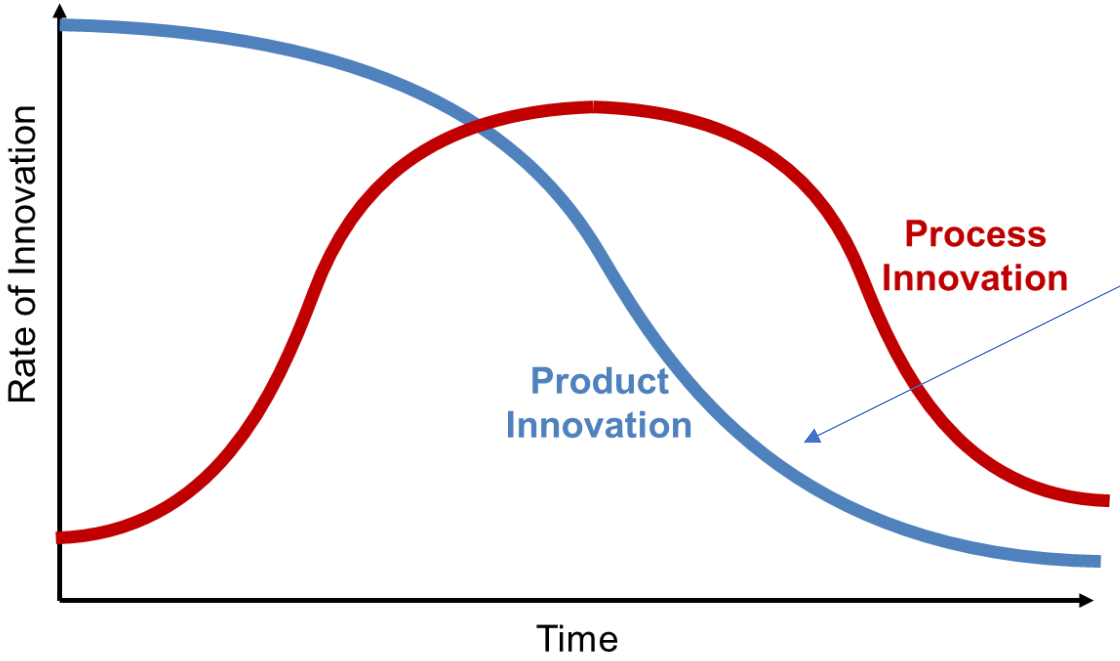
of the industry life cycle, before dominant design: Product innovation is the strategic imperative



Late phase

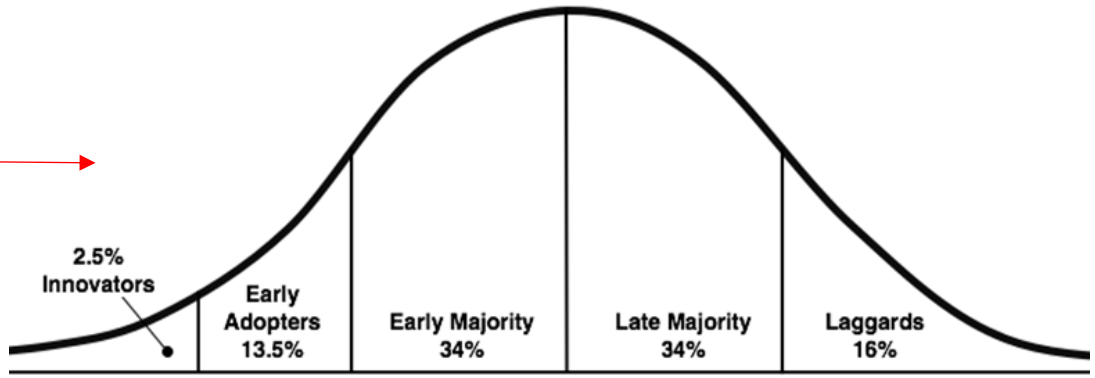
of the industry life cycle, after dominant design: Process innovation is the strategic imperative

INNOVATION over the industry lifecycle



Product innovation decrease along the time, because there is less possible innovations
And a winner standard (**Dominant Design**)

How customers adopt new innovations?

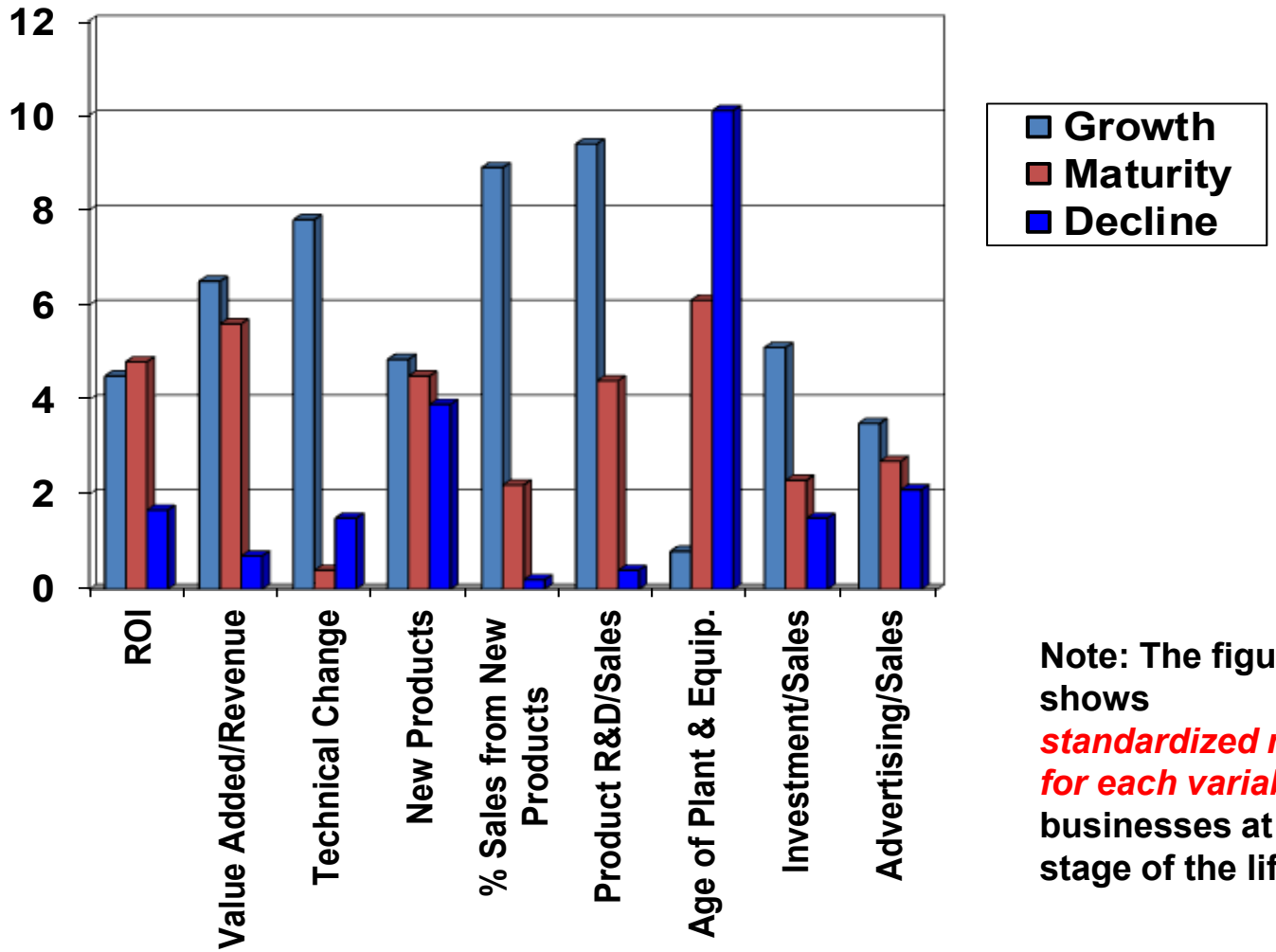


Do you think that

Life Cycle will affect the strategy?

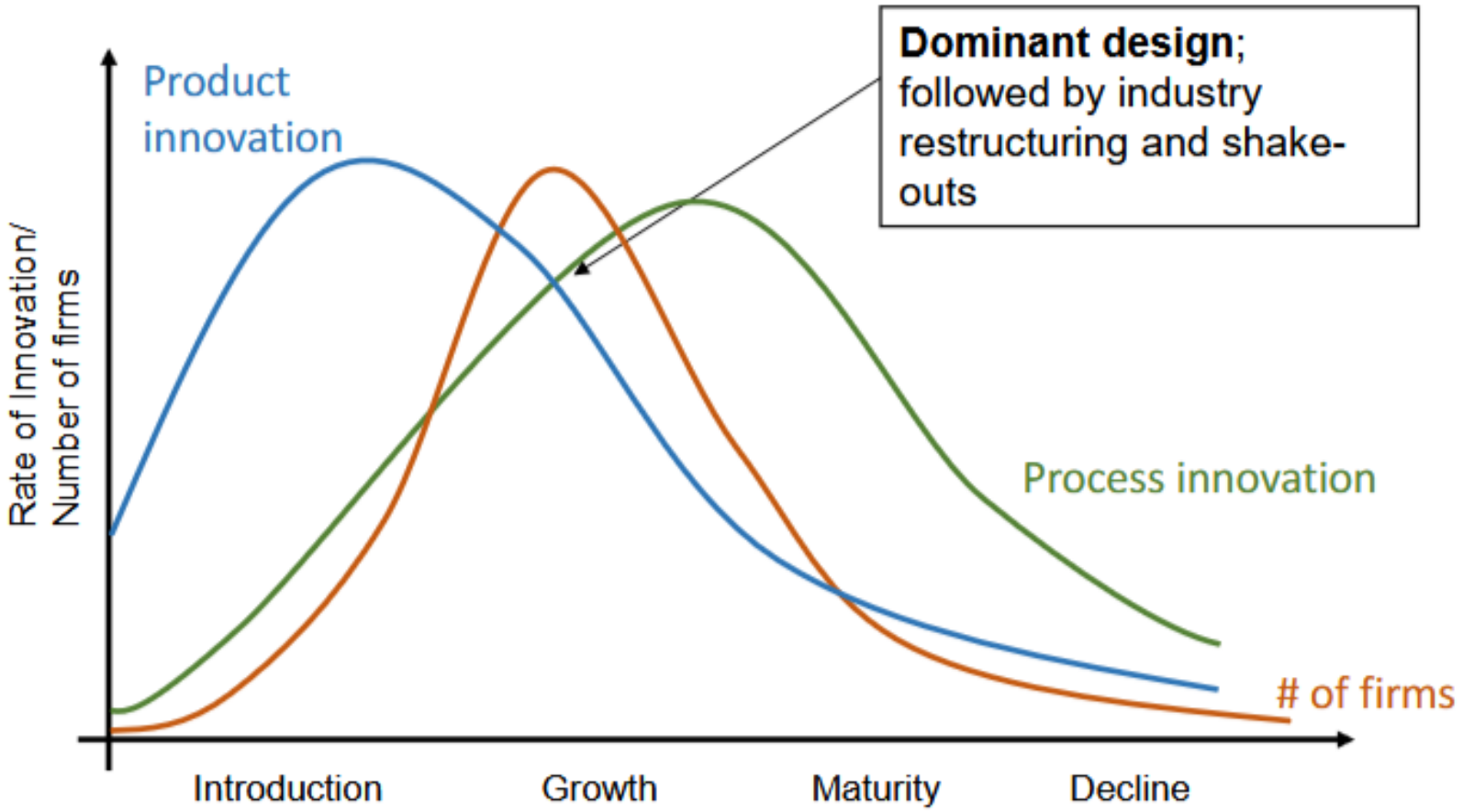
	INTRODUCTION	GROWTH	MATURITY	DECLINE
DEMAND	Early adopters	Rapid increase in market penetration	Replacement/ repeat buying; price sensitive customers	Obsolescence
TECHNOLOGY	Competing technologies; rapid product innovation	Standardization; rapid process innovation	Diffused know how; incremental innovation	Little innovation
PRODUCTS	Wide variety of features and designs	Design & quality improve; dominant design emerges	Commoditization; brand differentiation	Differentiation difficult
MANUFACTURING	Short-runs, skill intensive	Capacity shortage, mass-production	Over-capacity emerges; deskilling	Overcapacity
TRADE	------Production shifts from advanced to emerging countries------			
COMPETITION	Few companies	Entry, mergers exit	Consolidation	Price wars & exit
KSFs **Key success factors**	Product innovation Image of the firm and product	Process innovation Access distribution Brand building Fast Production Design for manu-facture;	Cost efficiency (scale economies, low cost inputs) Low Input costs Scale efficiency	Low margins & Overheads; Buyer selection; Rationalization capacity

PERFORMANCE across the Industry Life Cycle

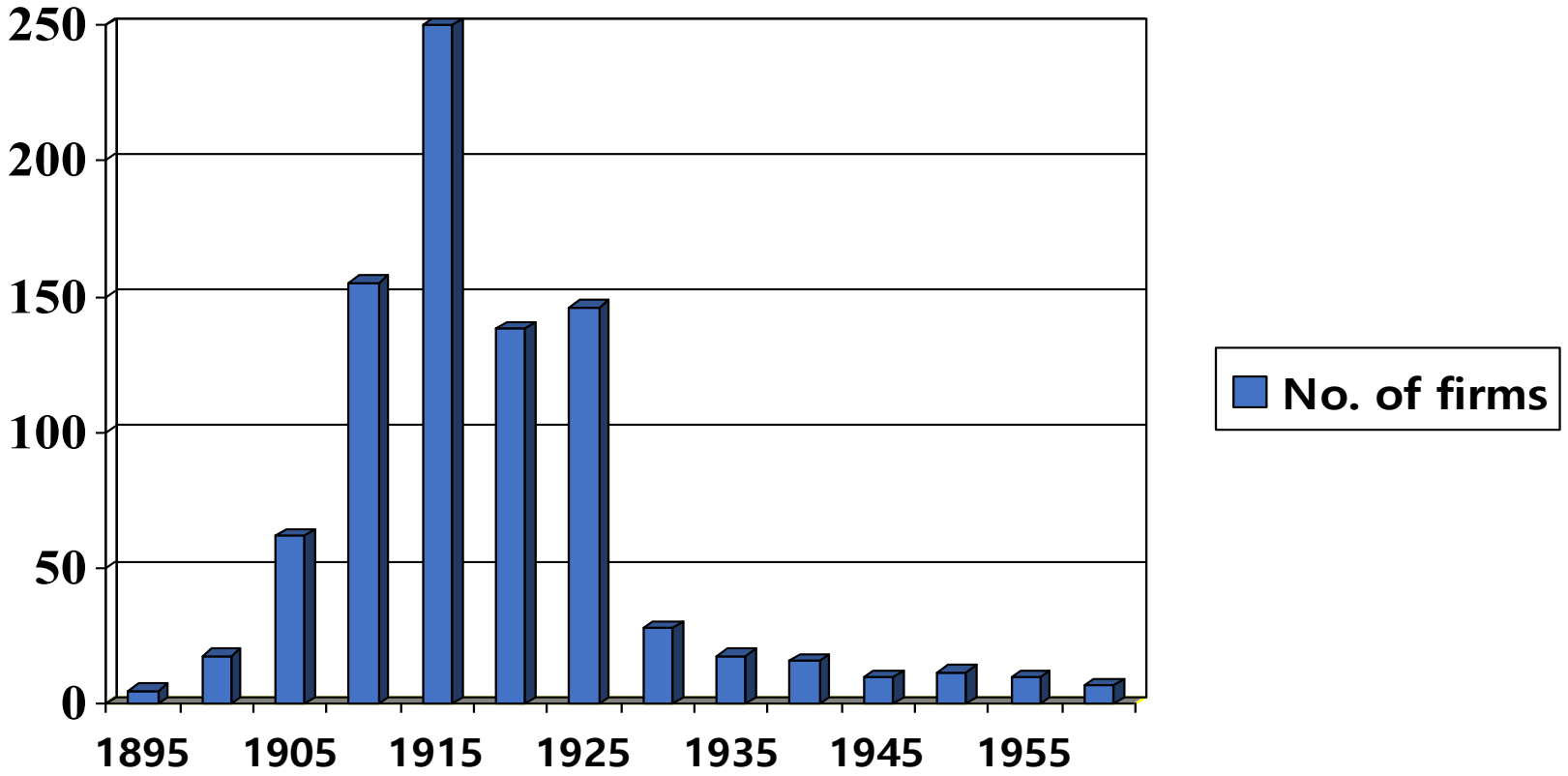


Note: The figure shows *standardized means for each variable* for businesses at each stage of the life cycle.

How does the number of firms in an industry develop over the industry lifecycle?



Number of Firms over the Industry Life Cycle: US Auto Industry 1895-1960



Source: S. Klepper, *Industrial & Corporate Change,* August 2002, p. 654.

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Life Cycle Industry Dynamics → Music Industry

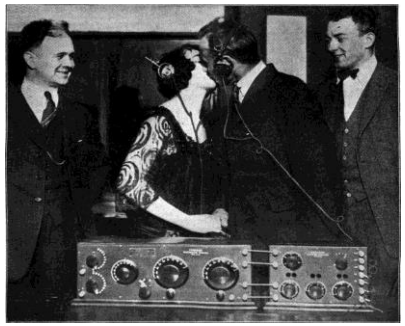
The Music Industry



1450s – print music



late 1800 – first phonographs



1920s– first radios



1950s – first LPs



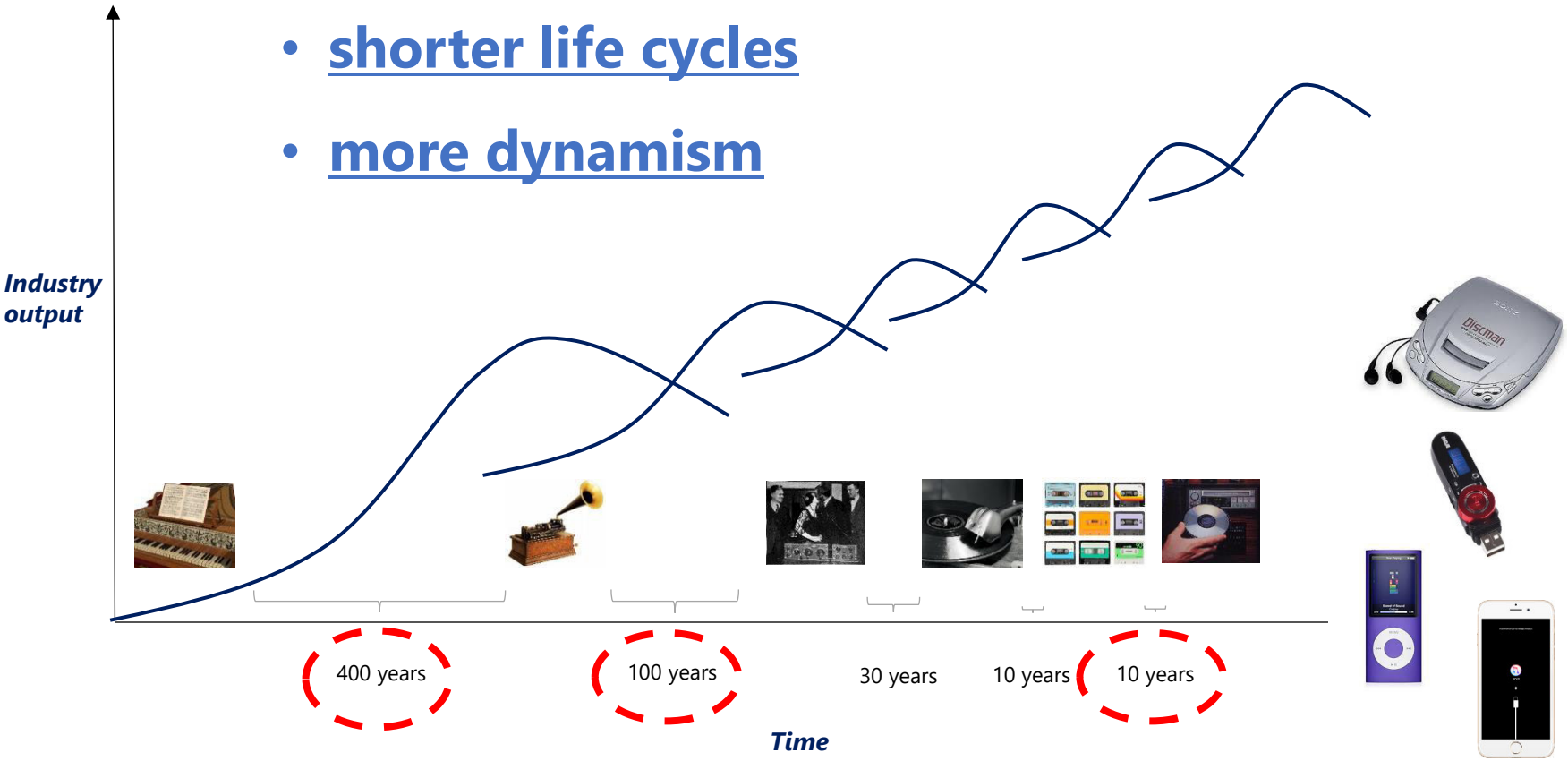
1960s – first cassettes



1970s – first CDs

Life Cycle Industry Dynamics → Music Industry

- New dominant designs
- shorter life cycles
- more dynamism

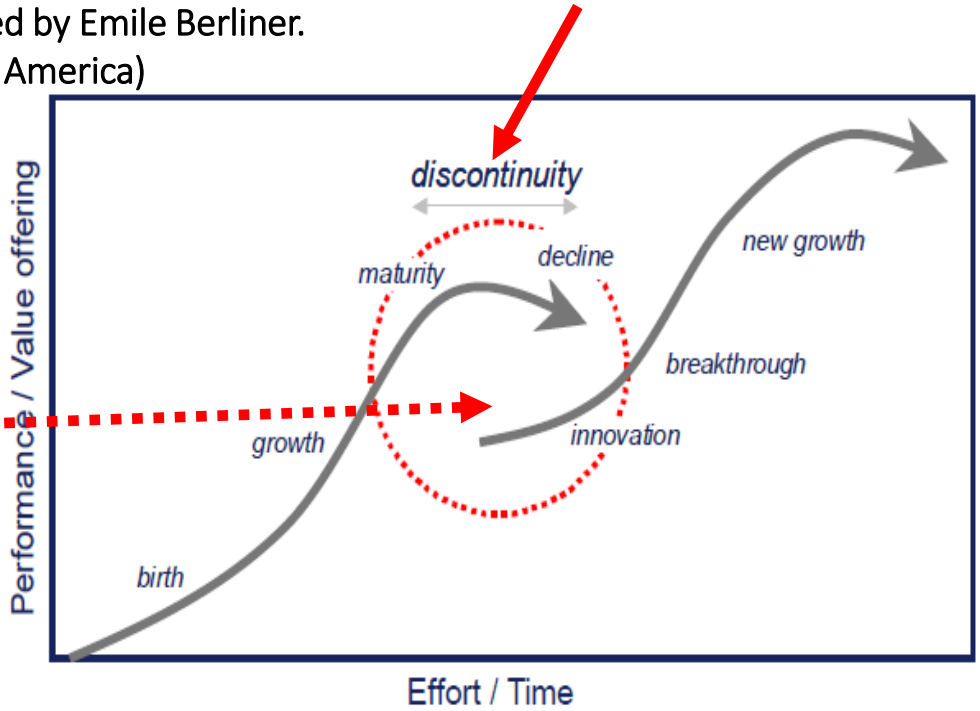


Continuity of the Life Cycle → Technological discontinuity

The Music industry and the Digitization

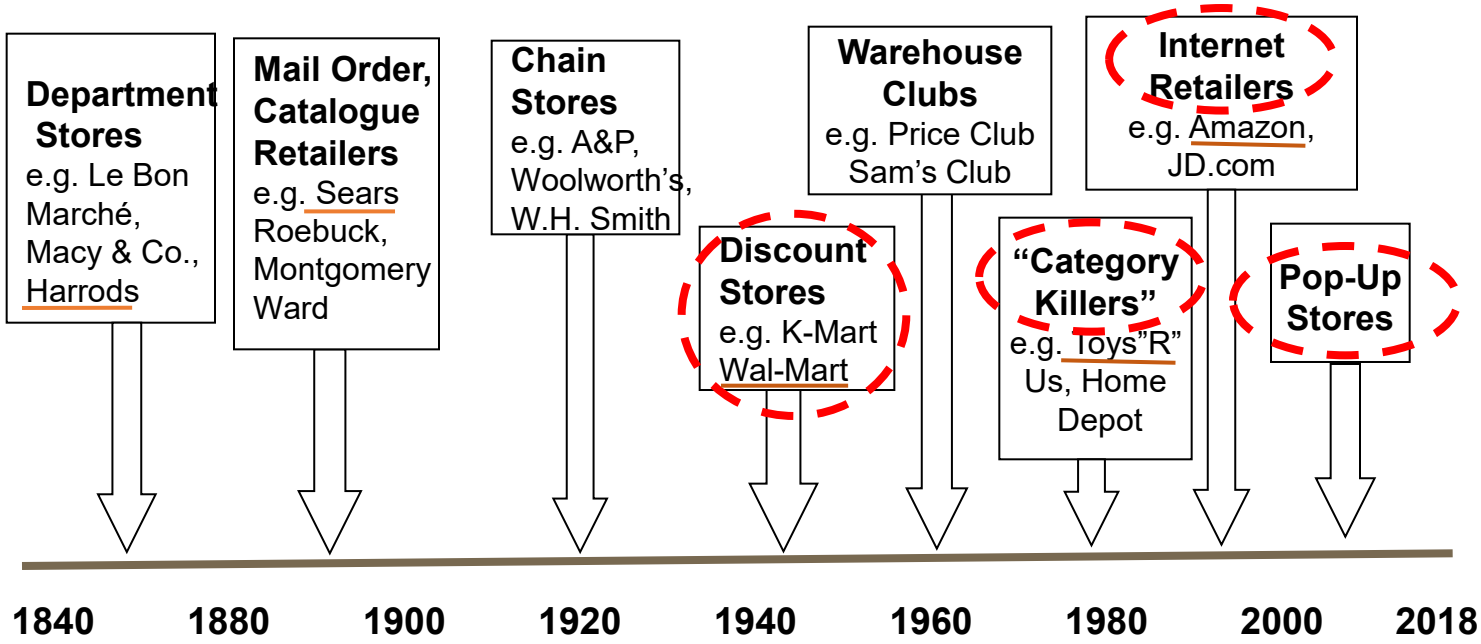
Technological discontinuity takes place

- 1888 – Berliner disk. The Gramophone invented by Emile Berliner.
- 1920´s- The Radio – RCA (Radio Corporation of America)
- 1950 –The LP
- 1950-1960 Rock & Roll. R&B (Rhythm & Blues)
- 1980 The CD,
- 1981 MTV.
- 1988 CDs surpasses LPs
- 1991 W.W.W. went to life to the world.
- 1993 MP3 Standard Published.
- 1999 Napster was founded by Shawn Fanning.
- 2003 iTunes Music store.
- 2005 YouTube
- 2008 Spotify.
- 2009 VEVO



Innovation and Renewal over the Life Cycle:

Retailing



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Factors of the Life Cycle Industry Dynamics

Factors that shape the competition over the evolution of the industry.

1. **EXTERNAL CHANGES** →

Exogenous shocks → Environmental external changes

E.g.: COVID vaccines; Electric car.

2. **DEMAND CHANGES** →

Changes in product demand (low/ high/ stable/ with different demographic trends)

E.g.: Low cost airlines; Fast food.

3. **INNOVATION CHANGES** (product or process) →

Creation, Production and Diffusion of Company Knowledge.

E.g.: Internet; Toyota processes; Ford continuous line production.

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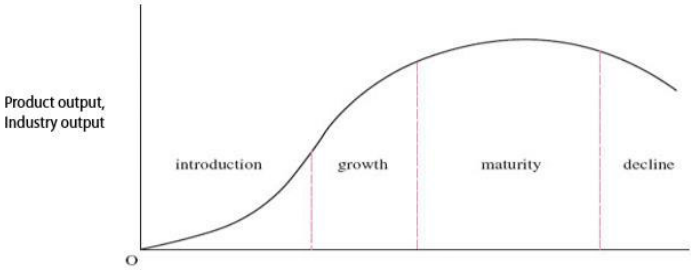
5. Competitive advantage in mature industries

How typical is the industry lifecycle pattern?

Specific pattern of the life cycle varies across industries…

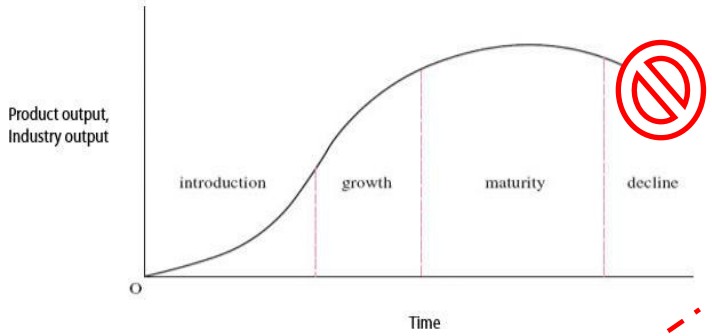
1. Time compression due to **fast innovation**:

e.g. Digital technologies industries



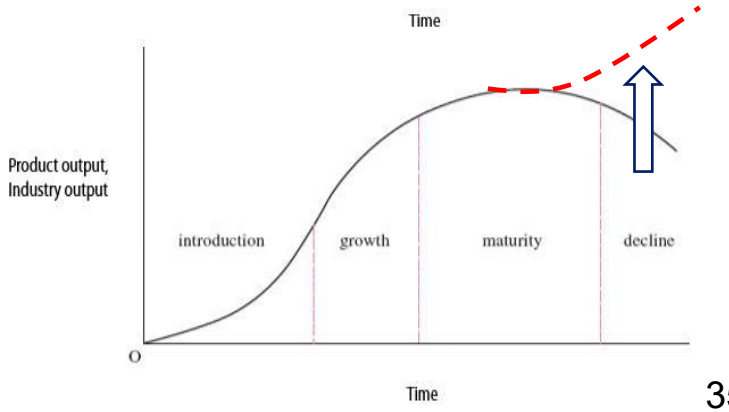
2. **No decline in basic needs**:

Providing basic necessities, they reach maturity, but not decline. e.g. food processing, construction, apparel.



3. **Convergence with other industries**:

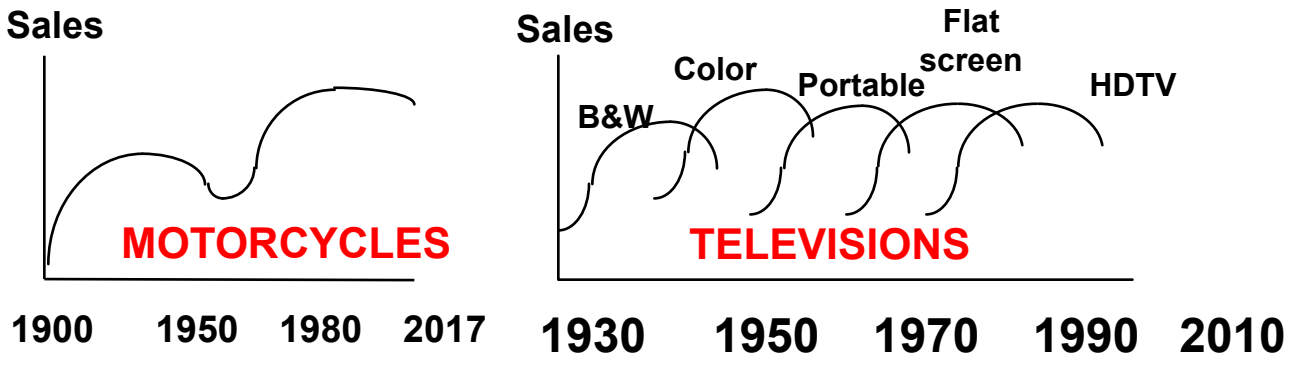
e.g. Cameras & Smartphones industries



4. Technology-intensive industries → **may retain features of emerging industries.**

e.g. pharmaceuticals, semiconductors, computers

5. Industries experience life cycle regeneration. e.g. motorcycles, TVs…

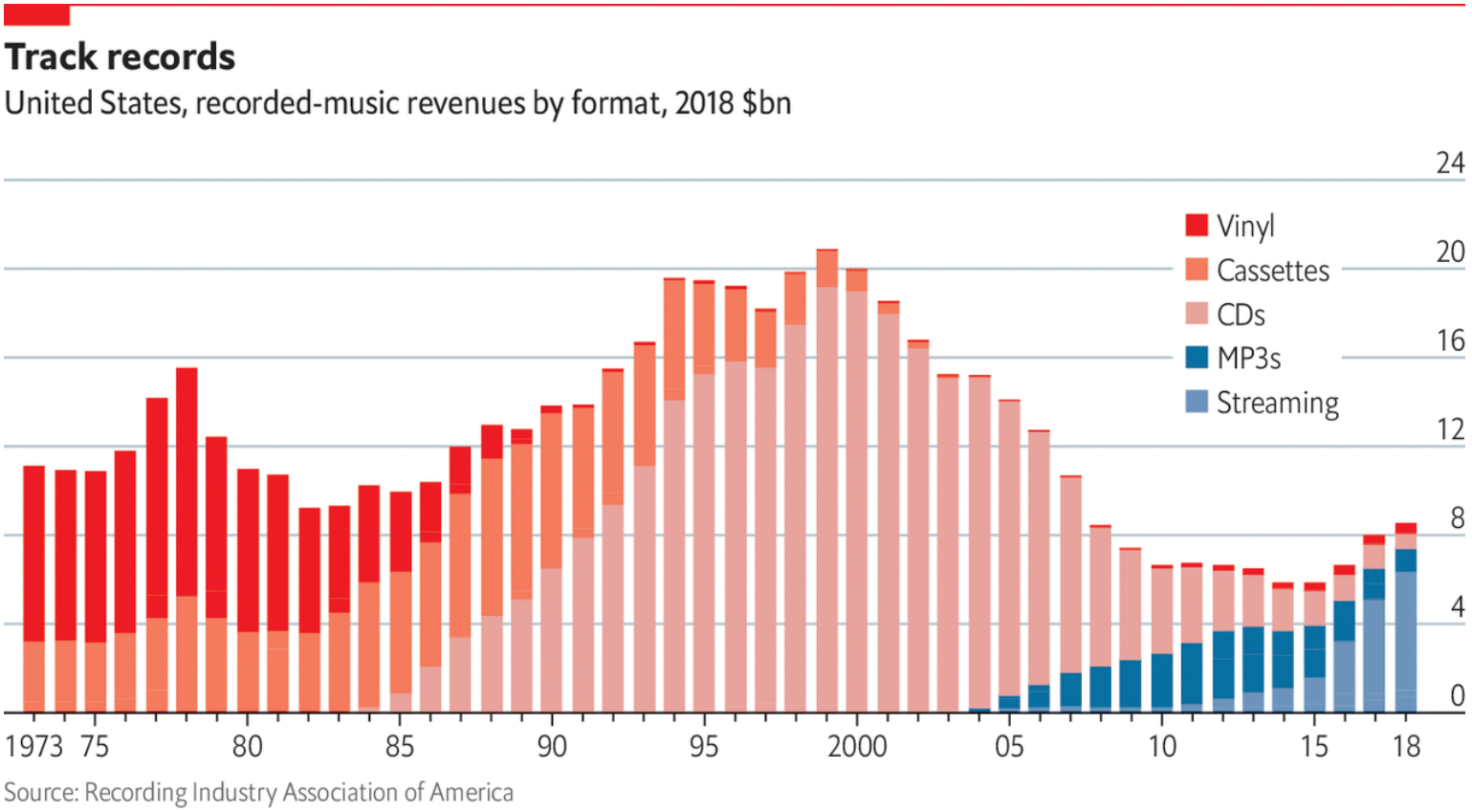


Life cycle model can help us to anticipate industry evolution, but dangerous to assume any common, pre-determined pattern of industry development.

Industries can experience revival

Revival of vinyl records

2019 sales will overtake those of fast-fading CD



1. The industry life cycle

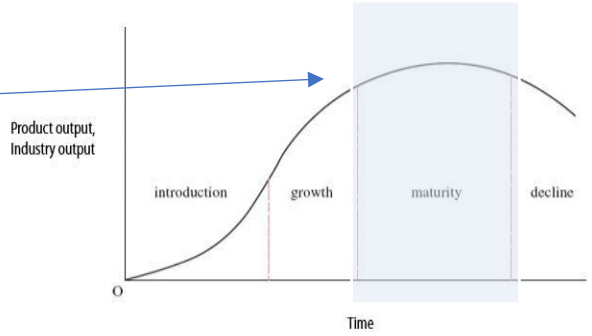
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5. Mature industries: Competitive advantage

Strategy Implementation in Mature Industries



Lean Manufacturing reducing costs and enhancing quality (total quality control)

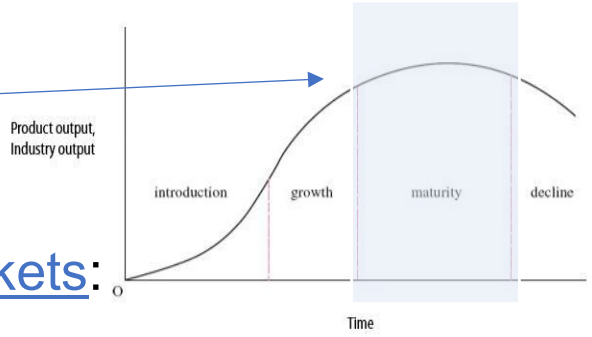
Powerful **global distribution** network and diversified portfolio of brands

Leverages on **economies of scale**, efficient supply chain and sustained low prices

Focus on **vertical integration**, owning mines on controlling raw material costs

Advanced logistic systems that allow for just-in-time delivery and sustainable operations

Strategy Implementation in Mature Industries



It is challenging to be profitable in mature markets:

- Overcapacity and Commoditization increase competitive pressure
- Difficult to establish and sustain competitive advantage
  - Less scope for Differentiation advantage (informed buyers, product standardization, lack of technological change)
  - Cost advantages are difficult to obtain and sustain (high diffusion of process technology = Firms become homogeneous)
  - Easier for new entrants to attack established firms (highly developed industry infrastructure together)

SOURCES OF COMPETITIVE ADVANTAGE IN MATURE INDUSTRIES

1. COMMODIZATION → means that cost efficiency becomes the primary source of competitive advantage

2. CORPORATE RESTRUCTURING → Periods of structural and strategic change

3. FURTHER SEGMENTATION & CUSTOMER SELECTION → Try to find your competitive niche

4. DIFFERENTIATION → Focus on being different and unique

5. STRATEGIC INNOVATION → Blue-ocean strategy, Value chain, reconfigurations, Redefining markets and market segments

SOURCES OF COMPETITIVE ADVANTAGE IN MATURE INDUSTRIES

1. **COMMODIZATION** → means that cost efficiency becomes the primary source of CA:
   - Economies of scale (facilitated by standardization)
   - Low-cost inputs
     - Migration and outsourcing to newly industrialized countries
     - Established firms can become locked into high salaries, inefficient working practices
   - Low Margins and overheads.

2. **CORPORATE RESTRUCTURING** → (= periods of structural and strategic change)

   Necessary when cost inefficiencies

   e.g.: Outsourcing, headcount reduction, and downsizing

3. FURTHER SEGMENTATION & CUSTOMER SELECTION:

- Even unattractive industries may offer **attractive niche markets** with strong growth of demand, few competitors, and potential for differentiation
- *(e.g.: Walmart's focus on rural areas in the 70´s)*
- Segmentation down to the level of specific customers and improved customer relationship management (CRM) → **Customer centricity**

4. DIFFERENTIATION



- Differentiation via **complementary services /products**

  (e.g.: Financing services, warranties, after sales service)
- Shift from physical differentiation to **Focus on image differentiation**
- Standardization constrains differentiation, but does not eliminate potential for differentiation

5. STRATEGIC INNOVATION

1. Strategic innovations: Blue-ocean strategy, Value chain reconfigurations, Redefining markets and market segments. Strategic Innovation requires **Developing new Capabilities**

- Create them internally → Difficult, due to myopic inertia, procedural inertia, size, power struggles
- Create them externally → Often better, through spinoffs or acquisitions

2. Embracing *new customer groups* (e.g. Wii vs PlayStation)

3. Augmenting, bundling & theming (Bundling additional products/services with the core offering, *e.g. Starbucks cafes in Barnes & Noble stores)*

4. Customer solutions (Integrated bundle of products/support services that are offered as a *customized package – e.g. All inclusive holiday package*)

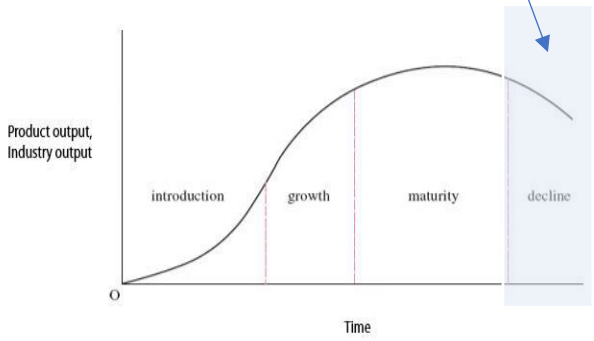
5. Culture of innovation → Liberation from the mature mind set (Maturity is a state of mind not a state of business) foster entrepreneurial organizational culture.

e.g. Telefonica → WAYRA & 42madrid

Strategy Implementation in Declining Industries

1. Adjusting Capacity to Decline Demand:

- Predictability of decline
- Barriers to exit
- Smooth exit for surviving firms



2. Alternative Strategies for Declining Industries:

- Grow → Leadership or Niche
- Decrease → Divest or Harvest

Strategic Transformation:

McKinsey Key success factors in a strategic transformation (5 mts)

https://www.youtube.com/watch?v=krywmXOaV6A



Porsche's Case – Questions to be solved

1. Why do you think Porsche launched two new products in 2013
(918 and Macan)? What were the reasons for the launch and the potential benefits that Porsche probably was hoping for?

2. Some experts argue that it has been a quite risky decision from
Porsche to launch the two products (and further launch them at the same time). Why? What may have been potential risks for Porsche when deciding the launch of the 918 and Macan?

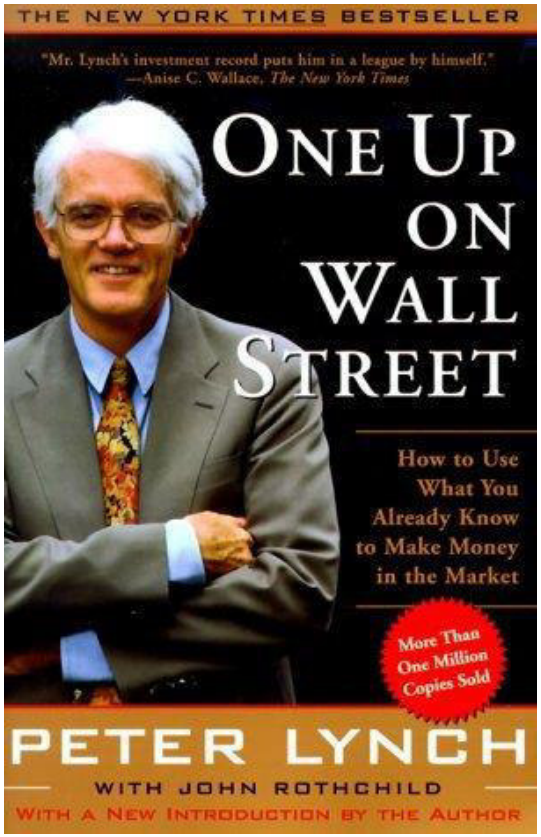
3. Was Porsche's decision to launch the 918 and Macan successful, in your opinion? Why?

Thanks a lot!

Any question?

Competitive advantage in mature industries

- The legendary investor, Peter Lynch (his fund at Fidelity averaged 30 percent annually for 15 years) has a philosophy of buying fast-growing companies in mature, 'lousy' (slow, or no-growth) industries.



Why do you think his investing strategy has been so successful?

ONE UP ON WALL STREET SUMMARY (BY PETER LYNCH) 15 m.
https://www.youtube.com/watch?v=s3jzbKW24jw

One up on Wall Street info audiobook (2h.)
https://www.youtube.com/watch?v=4t4DmRhcV5Y